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Clean Heating Solutions for Oil and Gas Wells in Low-Temperature Environments: Operational Performance and Techno-Economic Assessment of Multi-Configuration Air-Source Heat Pump Systems

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Against the backdrop of an accelerating global low-carbon energy transition, oil and gas field enterprises as traditional energy producers are under immense pressure to reduce energy consumption and carbon emission intensity in their production operations. During hydrocarbon extraction, particularly in gathering, transportation, and treatment processes, continuous thermal energy input is essential to maintain fluid mobility, prevent freezing, and meet process requirements. The resultant energy consumption constitutes a significant component of both operational costs and carbon footprints. Consequently, effectively reducing energy consumption in gathering/processing systems while achieving clean heating alternatives has emerged as one of the key imperatives for oilfield enterprises pursuing green.

Western China's desert regions harbor abundant oil and gas resources, yet their extreme environmental conditions impose severe challenges on production equipment. Taking this oilfield as a representative case, its location deep within a typical desert basin subjects operations to ground temperature extremes ranging from −33.2°C to 70.6°C. Under such conditions, extracted fluids require continuous heating at well sites to ensure safe and uninterrupted transportation. Historically, gas-fired heaters and electromagnetic heaters have served as primary heating solutions, but the former suffers from fossil fuel consumption and direct carbon emissions, while the latter incurs prohibitively high electricity demands. To support China's dual-carbon goals and advance clean energy adoption at well sites, heat pump technology has emerged as a promising alternative due to its utilization of ambient thermal energy, high efficiency, and low carbon footprint.[1] Consequently, the oilfield has deployed nearly 100 heat pump units across multiple configurations, including single-stage, cascade, transcritical CO_2, and solar collector-integrated systems, aiming to replace conventional heating methods. However, the actual operational performance and reliability of air-source heat pumps under frigid conditions remain inadequately validated.[2] The absence of systematic field data and in-depth analysis has created significant uncertainty, severely constraining large-scale implementation of this technology in similar harsh-environment oilfields.

To address this research gap and scientifically evaluate the applicability and energy efficiency of various heat pump systems under extreme cold conditions in actual well sites, this study implemented systematic field monitoring and analysis of seven representative heat pump systems configurations during a typical low-temperature season. The investigated systems encompassed single-stage, cascade, transcritical CO_2, and flat-plate collector-integrated units. Throughout the winter operational period (November 2024 to March 2025), the research team collected 3000 high-resolution datasets capturing critical parameters including flow rates, inlet/outlet temperatures, partial composition and ambient conditions. Applying the first law of thermodynamics and standardized heat transfer models, we quantitatively evaluated three key performance indicators: actual heat delivery capacity, coefficient of performance (COP), Clean replacement rate. Through comprehensive techno-economic analysis considering capital expenditure and operational outcomes, this research identifies optimized solutions for clean thermal energy supply in hydrocarbon fields under analogous extreme environments.

Introduction

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Geographical Location and Climate Characteristics of the Oilfield

The Tarim Oilfield is situated in the Tarim Basin in China, characterized by a typical arid climate and unique geographical environment. Geographically, the basin spans approximately 560000 square kilometers, exhibiting an irregular diamond shape and surrounded by mountains on three sides. The terrain slopes from west to east, with elevations ranging between 800 and 1300 meters. Climatically, the region experiences a typical warm-temperate continental arid climate, marked by abundant solar and thermal resources, with annual sunshine duration reaching up to 3000 hours. Temperature fluctuations are extreme, with the highest ground temperature recorded at 70.6°C and the lowest at −33.2°C. The diurnal temperature variation is significant, with an average daily range of 14-16°C and a maximum daily range of up to 25°C. These distinctive geographical and climatic conditions provide favorable natural endowments for oil and gas resource development, while simultaneously posing significant challenges to oilfield production equipment and process technologies.

The Application Situation of Heat Pumps

Compared to traditional electromagnetic heating technology, heat pump systems have become a key clean heating technology promoted by the Tarim Oilfield due to their higher coefficient of performance (COP) and significant energy-saving advantages. As of 2024, the oilfield has cumulatively installed nearly 100 heat pump of various types, primarily categorized into seven categories, with a total installed capacity of 11025 kW. In 2024, the total heat pump capacity equivalent (heating capacity) was 7966 tce, with an actual heat pump utilization volume (actual output) of 4035 tce. This is equivalent to saving 3.03 million cubic meters of natural gas and reducing carbon emissions by approximately 10491 tons annually.

Data Collection and Processing Methods

This study selected the period from November 2024 to March 2025 as the data collection period. First, this period fully covers the winter operating cycle of the Tarim Oilfield, enabling a comprehensive reflection of the operational characteristics of the heat pump system under extreme low-temperature conditions. Second, the period from November to March of the following year is the peak season for heating demand at the oilfield, with the most representative variations in system load. In terms of data collection, the research team continuously tracked and recorded the operational parameters of seven types of heat pump units. The primary parameters collected include: (1) ambient temperature, (2) inlet and outlet fluid temperatures of the heat pump, (3) circulation medium flow rate, (4) system electricity consumption and other energy consumption indicators. Approximately 3000 valid data samples were obtained throughout the winter. For raw data processing, this study employed the following methods. First, quality control was performed on the collected instantaneous values to exclude data points from abnormal operating conditions. Second, daily average values were calculated to eliminate the impact of short-term fluctuations. Finally, monthly statistics were compiled to establish a benchmark for evaluating the long-term operational performance of the system. This hierarchical processing method retains the dynamic characteristics of system operation while avoiding the interference of instantaneous fluctuations on overall performance assessment, providing a reliable data foundation for subsequent energy efficiency analysis.

Experimental Results and Analysis

Water Source Heat Pump

Water Source Heat Pump (WSHP) technology utilizes a small amount of electricity to upgrade low-grade thermal energy from surface or shallow water sources into usable medium or high temperature heat, achieving a COP of approximately 5. Compared to direct electric heating, this reduces energy consumption by approximately 70%.[3]

The HSL Processing Plant employs a WSHP system to replace conventional gas boilers for space heating and domestic hot water supply, with a designed thermal load of 2900 kW and supply/return water temperatures of 65/45 °C. This WSHP extracts waste heat from oily wastewater through a two-stage indirect heat exchange process utilizing an intermediate circulating water loop. Under rated conditions, the designed heating capacity of the WSHP system is 3342 kW, with an input power of 573 kW, yielding a COP of 5.83. Operational data for the HSL Processing Plant WSHP system are presented in Table 1. During the period from November 2024 to March 2025, the system operated with an average inlet temperature of 55.54 °C and an average outlet temperature of approximately 61.33 °C. The mean daily ambient air temperature was −0.81 °C and minimum temperature of −28.61 °C. The average input power was 228.46 kW, while the average heating output was 1012.4 kW. Based on these measurements, the calculated average COP was 4.41, and the clean energy replacement rate reached 76.48%. The heating capacity and COP of the water source heat pump system were lower than the design values. Analysis revealed that the primary reason for this performance degradation was increased pressure drop due to fouling in the oily wastewater/intermediate water plate heat exchanger.

Table 1—Operational Data Analysis of HSL Water Source Heat Pump

Month	Average import temperature (°C)	Average outlet temperature (°C)	Input power(kW)	Average temperature(°C)	Heating power(kW)	Average COP	Clean replacement rate (%)
2024/11	54.48	59.77	267.3	-0.47	1248.5	4.67	77.86%
2024/12	55.44	60.58	244.0	-12.71	1043.1	4.27	76.61%
2025/1	58.66	62.47	198.3	-12.44	628.8	3.17	68.46%
2025/2	56.0	62.61	224.6	6.25	1032.76	4.60	78.25%
2025/3	53.10	61.23	208.1	15.32	1108.8	5.33	81.24%
Average	55.54	61.33	228.46	-0.81	1012.4	4.41	76.48%

Cascade Air Source Heat Pump

Air Source Heat Pumps (ASHP) extract thermal energy from ambient air and transfer it to meet heating demands.[4] Their primary advantages include wide applicability and relatively low operating costs. However, they are susceptible to frost formation, and their efficiency diminishes significantly under low-temperature conditions. Cascade Air Source Heat Pumps (C-ASHP), a specialized type of ASHP illustrated in Fig. 1, employ a two-stage cascade refrigeration cycle.[5] Two independent refrigeration cycles are connected in series through an intermediate refrigerant-to-refrigerant heat exchanger. This cascade system architecture absorbs heat from cold ambient air in a sequential manner, thereby generating high-temperature hot water. Typical outlet water temperatures reach 75°C, with maximum outputs up to 95°C, while maintaining stable operation down to ambient temperatures of −35°C.

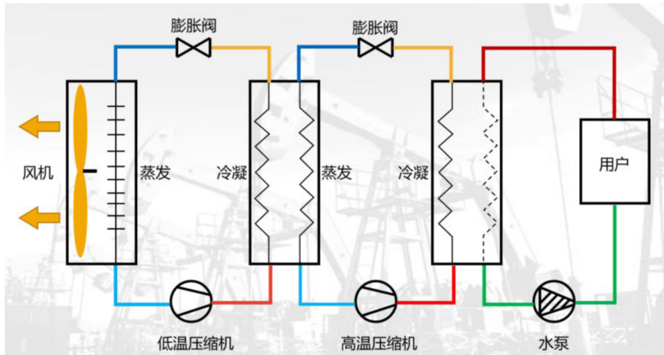

Figure 1—C-ASHPs cycle process

A total of 22 cascade air source heat pumps (C-ASHPs) have been deployed in the Oilfield, with a combined heating capacity of 1610 kW. These systems primarily serve wellhead heating and facility space heating applications. Performance data under low-temperature operation, detailed in Appendix Table 1. Actual COP values of C-ASHPs remain stable within the range of 1.69 to 2.21, averaging 2.015, following the trend of gradual decline with decreasing ambient temperatures. Notably, performance varies significantly across different application scenarios. C-ASHPs used for single-well heating consistently exceed their design COP standards and achieve clean energy replacement rates above 50%, demonstrating significant energy-saving and environmental benefits. Conversely, units deployed for space heating at the BZ Processing Plant exhibit actual COP values below design expectations, indicating an area requiring focused optimization efforts. The following section presents an in-depth alanalysis of heating for BZ treatment plants as arepresentative case study.

BZ processing plant employs the C-ASHPs to replace traditional gas boilers for space heating and domestic hot water supply, with a designed heat load of 1399 kW and supply/return water temperatures of 70/55°C. This C-ASHPs has a designed input power of 635 kW, and COP of 1.6 at an ambient temperature of −20°C. Operational data recorded during low-temperature conditions are presented in Table 2. Based on the operational data, the cascade air-source heat pump system at the BZ processing plant exhibited its lowest average COP value of 1.38 in December 2024. The daily average ambient temperature throughout the month ranged between −15°C and −6°C, representing the most challenging operating conditions for the heat pump. This suboptimal performance may be attributed to the close spacing and parallel arrangement of the outdoor units, which likely led to mutual airflow interference and a resulting cold island effect. This configuration appears to have exacerbated performance degradation under low-temperature conditions, causing the COP to fall below the design value.

Table 2—Analysis of Low-Temperature Operating Data for the BZ Treatment Plant's C-ASHPs

Month	Average import temperature (°C)	Average outlet temperature (°C)	Average cycle flow(m3/d)	total power consumption(kWh)	Averge temperature (°C)	Heat gain (MJ)	Average COP	Clean replacement rate (%)
2024/11	50.83	57.38	2040	154231.70	2.00	1177671.60	2.12	52.72%
2024/12	52.53	59.21	2280	321068.30	-10.57	1599096.24	1.38	27.68%
2025/1	59.86	67.41	2280	166974.50	-5.89	939980.16	1.56	36.00%
Average	54.41	61.33	2200	214091.50	-4.82	1238916.00	1.69	38.80%

Transcritical CO_2 Air Source Heat Pump

Transcritical CO_2 Air Source Heat Pumps (TC-ASHP) belong to the ultra-low-temperature ASHP category, utilizing carbon dioxide as the refrigerant to transfer heat via a transcritical cycle.[6] These systems maintain high efficiency at temperatures as low as −20 °C, deliver heating outputs exceeding 60 °C, and are ideally suited for space heating in cold regions, industrial processes, and renewable energy integration. At YM46-H4T well, a TC-ASHP system was implemented to replace conventional electromagnetic heaters for wellhead fluid heating. The well produces 78 t/d of fluid with a wellhead pressure of 0.64 MPa. Operating in direct heating mode, the oil-water-gas mixture enters the TC-ASHP condenser at 31 °C and is heated to 53 °C. Under the design condition of −7 °C ambient temperature, the system specifications include a heating capacity of 66 kW, an input power of 23 kW, and a COP of 2.83.

This CO_2 air-source heat pump is a new experimental device. Due to insufficient refrigerant injection, damaged axial fans, and substandard explosion-proof measures, it is still undergoing debugging during the heating season. The site relies on electric heaters to meet heating needs, and only partial heat pump test operation data has been collected. Manufacturer-collected test data from January 22-25 presented in Table 3 indicate operation under ambient temperatures ranging from −20°C to −3°C, with a mean ambient temperature of −10.7°C. Key performance metrics include average inlet and outlet fluid temperatures of 37.90 °C and 52.82 °C respectively, an average thermal load of 1155 kWh, and mean electricity consumption of 477 kWh. Derived calculations yield an average COP of 2.42 and clean energy replacement rate of 58.65%. Although the actual COP marginally underperformed against design specifications, the system demonstrated robust heating capability throughout extreme low-temperature conditions.

Table 3—YM46-H4T well TC-ASHP low temperature operationdata

Date	Average import temperature (°C)	Average outlet temperature (°C)	daily heat load(kWh)	daily power consumption(kWh)	Average COP	Clean replacement rate (%)
22/January	37.02	51.73	1139.66	506.75	2.25	55.53%
23/January	38.09	52.57	1121.23	461.52	2.43	58.84%
24/January	38.60	51.94	1033.08	406.18	2.54	60.68%
25/January	37.88	55.02	1327.57	537.0	2.47	59.55%
Average	37.90	52.82	1155.39	477.86	2.42	58.65%

CO_2 Cascade Air Source Heat Pump

The CO_2 cascade air source heat pump (CO_2-C-ASHP) is a high-efficiency heating device that integrates the advantages of CO_2 refrigerant with cascade cycle technology. The MS Apartment complex utilizes a CO_2 cascade air source heat pump system to replace traditional gas boilers and air conditioners for space heating and cooling. The system has a designed heating load of 319 kW, with heating supply/return water temperatures of 60/50°C, and a designed cooling load of 402 kW, with cooling supply/return water temperatures of 7/12 °C. At an ambient temperature of −20°C, this CO_2 cascade air source heat pump has a designed heating capacity of 408 kW and a power input of 186 kW, resulting in a COP of 2.19. The corrected operational data for the MS Apartment's CO_2 cascade air source heat pump system is presented in Table 4. During the heating season, the system exhibited an average inlet temperature of 49.81 °C, an average outlet temperature of 54.19 °C, and an average daily ambient temperature of −3.93 °C. The average monthly electricity consumption was 93558 kWh, with a heat output of 771205 MJ. The calculated average COP was 2.29, and the clean energy replacement rate was 57.49 %. The COP during low-temperature operation aligns well with the design COP.

Table 4—MS Apartment CO2-C-ASHP operational data

Month	Average import temperature (°C)	Average outlet temperature (°C)	Aaverage cycle flow (m³/d)	Total power consumption (kWh)	Averge temperature (°C)	Heat gain (MJ)	Average COP	Clean replacement rate (%)
2024/11	51.29	55.73	59.73	94780.80	-0.47	801820.66	2.35	57.42%
2024/12	51.03	55.09	59.97	97228.80	-12.71	761190.19	2.17	53.99%
2025/1	50.79	54.87	60.35	96902.40	-12.44	768550.61	2.20	54.58%
2025/2	50.17	54.94	59.42	83598.48	-4.04	685171.87	2.28	62.33%
2025/3	45.78	50.32	59.17	95281.60	10.00	839295.24	2.45	59.11%
Average	49.81	54.19	59.73	93558.42	-3.93	771205.71	2.29	57.49%
Total	/	/	/	467792.08	/	3856028.57	/	/

According to the Table 4, the CO_2-C-ASHP of MS Apartments recorded its lowest average COP value in December 2024, reaching only 2.17. Therefore, an analysis of its December operational data is presented in Figure 2. Overall, the daily average ambient temperature showed a declining trend, dropping from around −5°C to approximately −19°C, while the average COP value of the heat pump fluctuated between 2.0 and 2.2. Notably, the lowest daily average temperature was −19°C, at which point the heat pump's average COP value was 2.01. It is not significantly different from the designed value and exhibits excellent performance under low-temperature conditions.

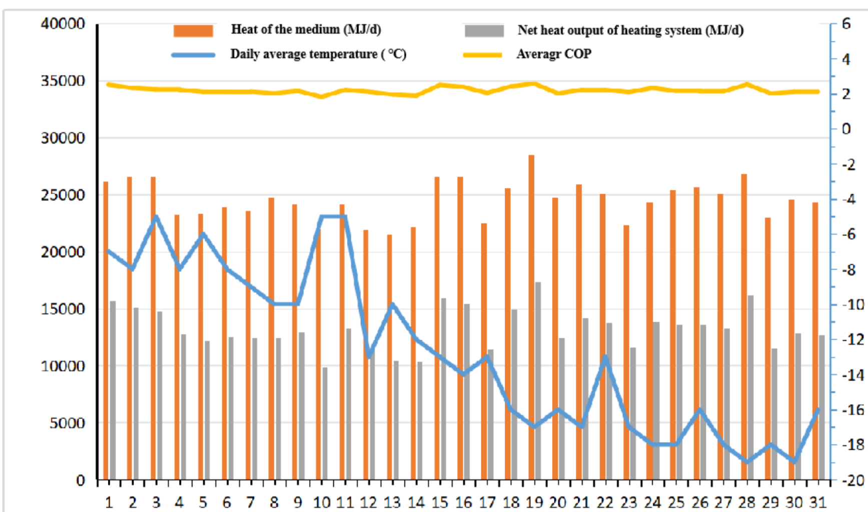

Figure 2—MS Apartment CO2-C-ASHP Operational Data for December 2024

Combination Heat Pump

A combination heat pump (CHP) refers to a system that couples the cycles of two or more working fluids and integrates a thermal storage system to achieve a greater temperature lift, without direct connection between the cooler of the low-temperature cycle side and the evaporator of the high-temperature cycle side.[7] Combination heat pumps typically operate in multiple modes such as high-temperature mode and low-temperature mode to adapt to different operating conditions. A total of 15 combination heat pumps have been deployed in the oilfield, delivering a combined heating capacity of 1410 kW primarily for apartment heating. Specific low-temperature operational performance data is shown in Appendix Table 1. The actual COP values under all three scenarios were lower than design specifications, indicating suboptimal heating performance. The following analysis will examine operational details using the YH2

Apartment combination heat pump as a case study. Designed for −7°C ambient conditions, it achieves 470 kW heating output at 186.5 kW input power with a COP of 2.52. Operational data in Table 5 shows that from November 2024 through March 2025, average inlet temperature was 39.72°C, average outlet temperature approximately 43.26°C, and average daily ambient temperature 4.1°C. Total electricity consumption reached 462194 kWh while heat output was 4003714 MJ, yielding a calculated average COP of 2.4 and clean energy substitution rate of 58.42%.

Table 5—YH2 Apartment CHP low temperature operating data

Month	Average import temperature (°C)	Average outlet temperature (°C)	Average circulation flow rate (m3/d)	Total power consumption (kWh)	Average d temperature (°C)	Heat gained (MJ)	Average COP	Clean replacement rate (%)
2024/11	37.82	39.93	1923.30	52461.28	7.15	501859.47	2.66	62.35%
2024/12	46.34	50.53	1751.92	118596.80	5.29	990610.45	2.32	56.88%
2025/1	46.76	51.48	1794.68	143340.80	-2.82	1095854.76	2.12	52.88%
2025/2	41.13	44.88	1770.83	88369.76	-1.60	780878.03	2.45	59.24%
2025/3	26.53	29.47	1716.67	59425.60	12.00	634510.80	2.96	66.27%
Average	39.72	43.26	1791.48	92438.85	4.00	800742.70	2.40	58.42%
Total	/	/	/	462194.24	/	4003714	/	/

According to the Table 5, the combination heat pump at YH2 Apartment also recorded its lowest average COP value in January 2025, reaching only 2.12. Consequently, an analysis of its January operational data is presented in Fig. 3. The daily average ambient temperature generally exhibited a declining trend, with the lowest daily average temperature reaching −10°C. Meanwhile, the system's daily average COP value fluctuated between 1.6 and 2.6, showing a decreasing trend as the ambient temperature dropped.

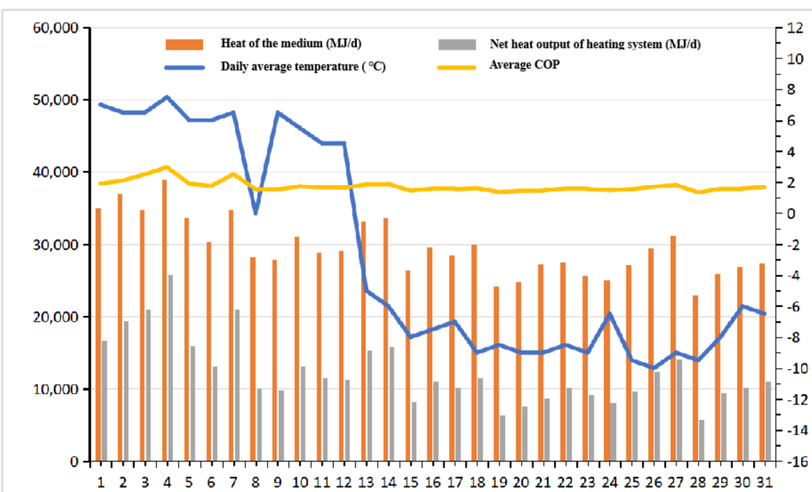

Figure 3—YH1 Apartment CHP operating data for January 2025

Flat-Plate Collectors + Single-Stage Air Source Heat pump

DN Apartment utilizes an flat-plate collectors + single-stage air-source heat pump (FPC+SASHP) to replace traditional gas boilers for domestic hot water supply. The designed daily hot water supply capacity is 40 tons/day, with a make-up water temperature of 12°C and supply temperature of 60°C. The system comprises 3 air-source heat pump outdoor units (refrigerant R410a), one 40 m^3 thermal storage water tank, and 30 arrays of flat-plate collectors (15.02 m^2 each). Designed for −7°C ambient temperature, it delivers a heating capacity of 186 kW with an input power of 45.6 kW, achieving a COP of 4.08. Operational data in Table 6 shows that during the heating season, the system maintained an average inlet temperature of 12.7°C, average outlet temperature of approximately 49.78°C, and average daily temperature of 4.02°C. Total electricity consumption reached 151864 kWh while heat output was 1045958 MJ, yielding a calculated average COP of 1.91 and clean energy substitution rate of 47.70%. Under low-temperature conditions, the actual average COP of DN Apartment's system was significantly lower than the design value. This is primarily due to two factors: First, winter temperature drops and snowfall caused extensive ice formation or snow coverage on the collector surfaces. Concurrently reduced solar irradiation proved insufficient to melt the ice/snow, rendering the flat-plate collectors largely ineffective and forcing the air-source heat pump to provide most of the heating. Second, the single-stage compression process of this air-source heat pump, combined with observed severe frost buildup on the outdoor heat exchanger observed on-site, resulted in poor heating performance under low-temperature conditions.

Table 6—DN flat plate collector + single-stage air source heat pump low temperature operating data

Month	Average import temperature (°C)	Average outlet temperature (°C)	Average hot water consumption (m^3/d)	Total power consumption (kWh)	Average temperature (°C)	Heat gained (MJ)	Average COP	Clean replacement rate (%)
2024/11	19.59	51.39	44.43	20712.80	11.20	177968.70	2.39	58.08%
2024/12	14.40	49.30	44.48	34300.08	0.58	202088.71	1.64	39.34%
2025/1	10.03	47.84	45.10	38848.98	-1.24	222085.08	1.59	32.46%
2025/2	9.30	49.95	44.61	28422.40	5.54	213267.18	2.08	52.00%
2025/3	10.18	50.40	45.50	29580.00	16.0	230548.50	2.17	53.54%
Average	12.70	49.78	44.82	30372.85	4.02	209191.63	1.91	47.70%
Total	/	/	/	151864.26	/	1045958	/	/

Evacuated Tube Collectors +Single-Stage Air Source Heat Pumps

YH Apartment employs two evacuated tube collectors + single-stage air-source heat pumps (ETU+SASHP) to replace traditional gas boilers for domestic hot water supply. The designed daily hot water capacity is 30 m^3/d for YH1 apartment and 20 m^3/d for YH2 Apartment, with make-up water at 12 °C supplied at 60 °C. Each system configuration includes: four R410a refrigerant air-source heat pump outdoor units, one 50m^3 thermal storage tank, and fifty arrays of evacuated tube collectors (5.04m^2 per array). Designed for −7°C ambient conditions, the system delivers 248 kW heating output at 60.8 kW input power with a COP of 4.08. During low-temperature operation, the actual average COP of this evacuated tube collector system was significantly lower than design specifications, as shown as Appendix Table 1. The primary performance limitations mirror those observed in DN Apartment's flat-plate collector system.

Performance and Investment Analysis of Heat Pump Applications

Analysis of heating season operational data from seven heat pump systems reveals the following COP ranking under low-temperature conditions, ordered from highest to lowest: WSHP, TC-ASHP, CO_2-C-ASHP, CHP, C-ASHP, FPC+SASHP and ETU+SASHP. Notably, WSHP maintain exceptional performance in low-temperature environments with COP values reaching 4.41, as their operation remains largely unaffected by ambient temperature fluctuations due to stable heat sources. TC-ASHP demonstrate outstanding heating performance at −10.7 °C, achieving COP values of 2.42 under low-temperature operating conditions. Both evacuated tube collectors+single-stage air-source heat pumps and flat-plate collectors+single-stage air-source heat pump exhibit comparable performance, with COP values approximately within the 1.9-2.2 range-significantly lower than design COP. The investment in heat pump systems was statistically categorized according to two application scenarios: heating supply and single-well heating. The results are presented in Appendix Table 2, where the unit investment cost is defined as equipment investment divided by heating capacity. The results indicate that: For heating supply applications, the unit investment cost for water-source heat pumps and cascade heat pumps ranges approximately from 1400 to 1700 yuan/kW, which is significantly lower than that of other heat pump types. For single-well heating applications, conventional air-source heat pumps exhibit the lowest unit investment cost. The unit investment costs for other types are similar, at approximately 8000 yuan/kW. A comprehensive technical and economic evaluation of various heat pumps indicates that under low-temperature winter conditions in oilfield applications, TC-ASHP, CO_2 -C-ASHP, CHP, and C-ASHP demonstrate COP values within the range of 2.0-2.242 and achieve clean replacement rates of 50%-58.65%, with relatively minor performance differences. However, for heating supply applications, CO_2-C-ASHP exhibit significantly higher unit investment costs compared to CHP and C-ASHP. In single-well heating applications, TC-ASHP show unit investment costs similar to those of C-ASHP.

Conclusion and Reflections

Heat pump technologies exhibit significant diversity in types and adaptability. Comparative analysis of seven heat pump systems under winter low-temperature conditions reveals that WSHP deliver optimal performance with the lowest unit investment cost, while collector-integrated air-source heat pump systems demonstrate relatively inferior results. Four categories (TC-ASHP, CO_2-C-ASHP, CHP, and C-ASHP) maintain favorable heating performance with COP values ranging between 2.0 and 2.42. Considering technical applicability, investment costs, and practical oilfield constraints, WSHP, CHP, or C-ASHP are recommended for heating applications in environments where temperatures persistently exceed −10°C. For individual well heating in remote oil and gas production sites-characterized by harsher operating conditions (e.g., extreme temperatures below −25°C) and lower ambient temperatures transcritical CO_2-C-ASHP emerge as the preferred solution.

As heat pump technology continues to mature, its adoption across oilfields will inevitably expand. Nevertheless, persistent challenges in technical selection, operational maintenance, and systemic management necessitate comprehensive solutions. Critical recommendations include: establishing definitive heat pump selection criteria, refining operation and maintenance protocols, optimizing operational strategies, piloting emerging technologies(e.g., high-temperature air-source heat pumps, PV-direct-drive heat pumps), and cultivating specialized technical teams for design, management, and maintenance. These integrated measures will ultimately establish a robust heat pump management framework to ensure sustainable implementation.

Reference

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